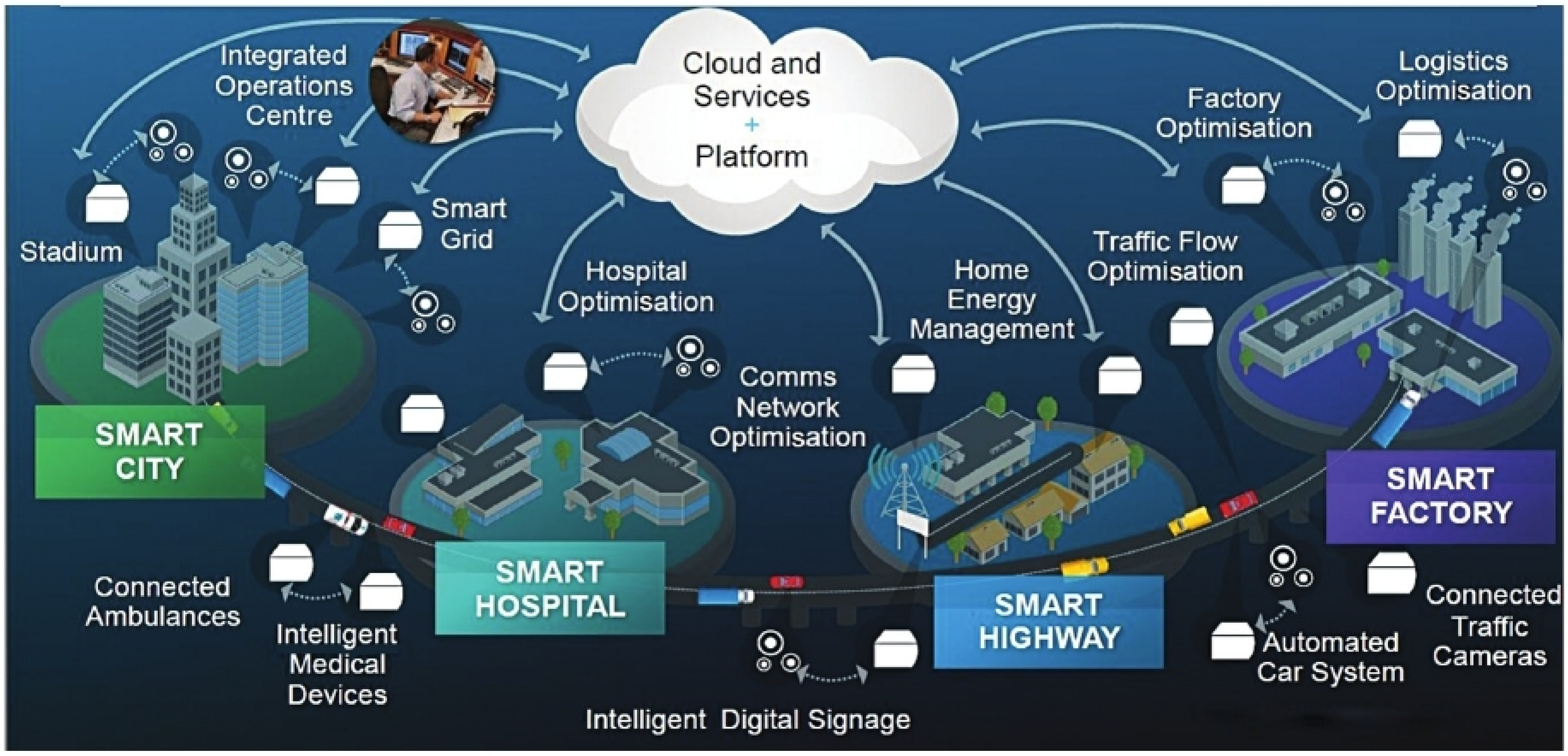

Connected utilities in a smart city (Credit: Cisco and/or its affiliates)

Second layer is the network layer that forwards IoT packets over reliable communication networks.

Third layer is the perception layer that gathers, captures, identifies and distinguishes information about the different objects in the physical world. This layer consists of sensors, cameras, RF tags, GPS, laser scanners and so on.

The IoT introduces new methods to capture live data and to send the same to a server or cloud. This helps with the monitoring of remote devices and decision-making based on remotely-captured information. The IoT products improve traffic management in cities, so that roads can become intelligent and recognise the number of vehicles on them.

To use a particular technology, it is important to understand and identify the requirements of that technology. The IoT platforms connect multiple heterogeneous systems in a city. For example, pediatric wards of hospitals in a city can be connected to schools to get health data of children. And based on this data, grants can be distributed.

We have seen that e-commerce has become extremely successful in India because Internet services have become cheaper. A ₹ 5 phone recharge has created growth for many e-services. Similarly, cheaper IoT services can help create connected smart cities.

A city has multiple concrete structures that can be monitored by putting passive wireless sensors in them. Periodically, radio signals are emitted from building structures that can be accumulated by the city municipal corporation. The figure above shows a highly-connected future smart city.

The IoT helps keep a watch on water levels of dams, lakes, sewage and the like. Water leakage must be sensed immediately for proper usage of water.

Waste management has become a major problem in urban cities. In smart cities, IoT devices attached to garbage containers send alerts when they fill up and, accordingly, garbage collector trucks pick up these containers. LPWAN technology can be used for this purpose.

Wireless patient monitoring systems are needed in hospitals. Here, passive and active sensors can be placed on the patients to monitor their physiological parameters. The sensors remain with the patients till the time they are discharged from the hospital. Highly-secure IoT systems must be created in compliance to privacy laws.

A smart grid is needed in every city. Intelligent power controllers have already been designed, and smart e-meters are being used to read consumption of electricity. The IoT is already playing a great role in cost-effective power generation, distribution and transmission.

Companies like Google, Uber, Renault, Ford and Volvo are developing autonomous vehicles. Driverless cars are a key component of connected smart cities. Air taxis will soon become common in ultra-smart cities.

IBM Bluemix is a platform for Infrastructure as a Service (IaaS) for smart cities. AirVantage is a machine-to-machine (M2M) connectivity platform to connect IoT objects. Nokia and Cisco are developing many connected smart city IoT platforms. END

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